Statement of Work (SOW) for Data Collection, Analysis, and Reporting

1. Data Acquisition

- Obtain source data through one or more of the following methods:
  - Creating datasets manually
  - Loading data from existing files (e.g., CSV, Excel)
  - Web scraping to retrieve live or updated data from websites

2. Data Analysis and Processing

- Analyze and process the data using suitable tools:
  - Python (for scripting, automation, and advanced analysis)
  - Excel (for quick analysis, pivot tables, and basic visualization)
  - Power BI (for interactive dashboards and reports)
- Select the appropriate tool based on the task's complexity, data volume, and desired output.

3. Data Management and Version Control

- Save all modified or processed files as new versions to:
  - Maintain a record of modifications
  - Ensure data integrity and traceability
  - Facilitate comparison between different versions

4. Data Visualization and Final Reporting

- Use the processed data to create visualizations and reports through:
  - Python (e.g., Matplotlib, Seaborn, Plotly)
  - Power BI dashboards
  - HTML-based reports or web pages
- Deliver insights in an accessible and understandable format to stakeholders.

5. Ongoing Updates and Maintenance

- Establish a schedule for regular data updates:
  - Weekly, biweekly, or monthly
- Incorporate new data into existing datasets.
- Compare updated data with previous versions to identify trends or changes.
- Refresh reports and visualizations accordingly.

Deliverables:

- Raw source data files
- Processed and cleaned datasets
- Version-controlled files documenting modifications
- Visualizations and dashboards
- Updated reports following scheduled intervals

Tools & Technologies:

- Python (pandas, BeautifulSoup, matplotlib, etc.)
- Excel
- Power BI
- HTML/CSS/JavaScript (optional for web-based reports)

Note: Adjust the scope and tools as per project requirements and available resources.